MAGeCK Count Report

Author: Wei Li, weililab.org

Parameters

comparison_name is the prefix of your output file, defined by the “-n” parameter in your “mageck test” command. The system will look for the following files to generate this report:

- comparison_name.countsummary.txt
- comparison_name.count_normalized.txt
- comparison_name.log

```
# define the comparison_name here; for example,
# comparison_name='demo'
comparison_name='allnonorm'
```

Note: if the sample labels are too long, they may cause issues in some figures. If this is the case, set the following variable to “TRUE”, and a numbered label will be shown instead of the original label.

```
REPLACE_LABEL=FALSE
```

Preprocessing

Reading input files. If any of these files are problematic, an error message will be shown below.

```
cstable=read.table(count_summary_file,header = T,as.is = T)
nc_table=read.table(normalized_cnt_file,header = T,as.is = T)
```

Summary

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ readr     2.1.5
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.5.2     ✔ tibble    3.2.1
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.0.4
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

The summary of the count command is as follows.

Count command summary

File	Label	Reads	Mapped	Percentage	TotalsgRNAs	Zerocounts	GiniIndex
21062D-07-03_S104_L006_R1_001.fastq	T0	26739135	15452375	0.5779	13008	462	0.1028
21062D-07-04_S105_L006_R1_001.fastq	D7	30680921	17345990	0.5654	13008	2199	0.3762
01_subset_PSU/21062D-07-01_S102_L006_R1_001.fastq	UL	26434164	14712582	0.5566	13008	3563	0.4958
21062D-07-02_S103_L006_R1_001.fastq	LL	30520874	17278471	0.5661	13008	3928	0.6153

The meanings of the columns are as follows.

- File: The filename of fastq file;
- Label: Assigned label;
- Reads: The total read count in the fastq file;
- Mapped: Reads that can be mapped to gRNA library;
- Percentage: The percentage of mapped reads;
- TotalsgRNAs: The number of sgRNAs in the library;
- ZeroCounts: The number of sgRNA with 0 read counts;
- GiniIndex: The Gini Index of the read count distribution. Gini index can be used to measure the evenness of the read counts, and a smaller value means a more even distribution of the read counts.

If –day0label and –gmt-file options are provided, the following metrics will display the degree of negative selections of essential genes (provided by –gmt-file).

Count command summary

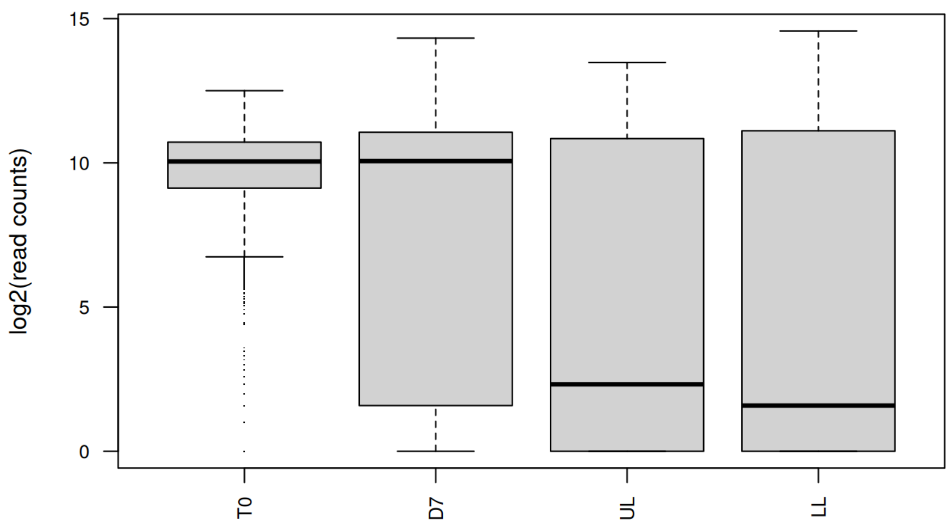
File	Label	NegSelQC	NegSelQCPval	NegSelQCPvalPermutation	NegSelQCPvalPermutationFDR	NegSelQCGen
21062D-07-03_S104_L006_R1_001.fastq	T0	0	1	1	1	[illegible]
21062D-07-04_S105_L006_R1_001.fastq	D7	0	1	1	1	[illegible]
01_subset_PSU/21062D-07-01_S102_L006_R1_001.fastq	UL	0	1	1	1	[illegible]
21062D-07-02_S103_L006_R1_001.fastq	LL	0	1	1	1	[illegible]

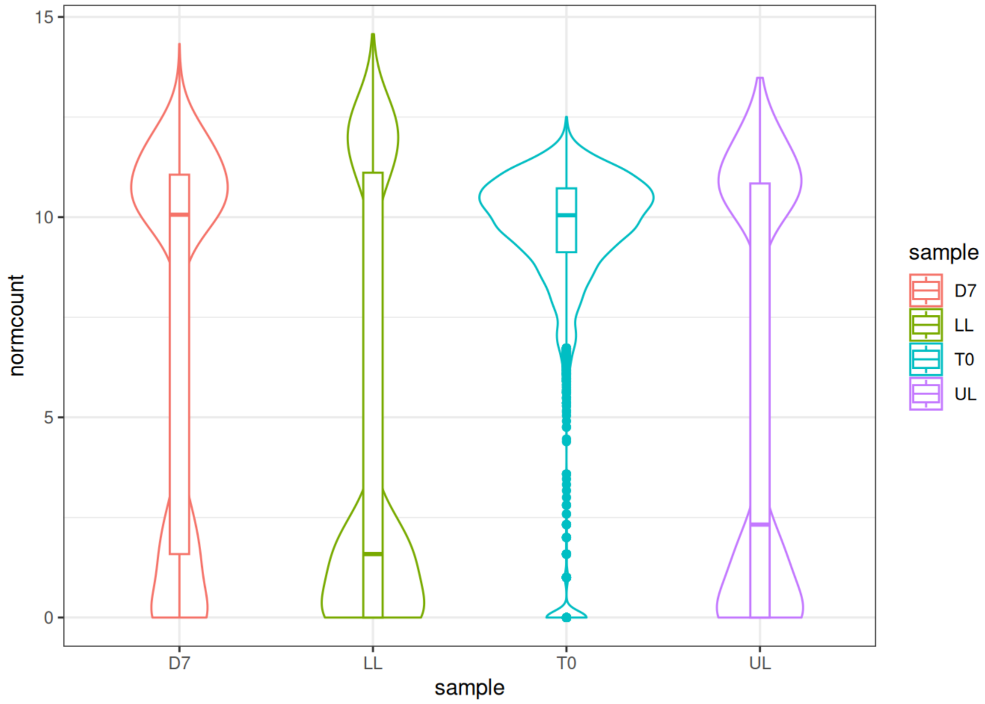
The meanings of the columns are as follows.

- NegSelQC: the enrichment score (ES) of essential genes in the negative selection list. The score is calculated using GSEA;
- NegSelQCPval: the associated p value of the enrichment score;
- NegSelQCPvalPermutation: the permutated p value of the enrichment score;
- NegSelQCPvalPermutationFDR: the adjusted permutated p value;
- NegSelQCGene: the number of genes used for the analysis.

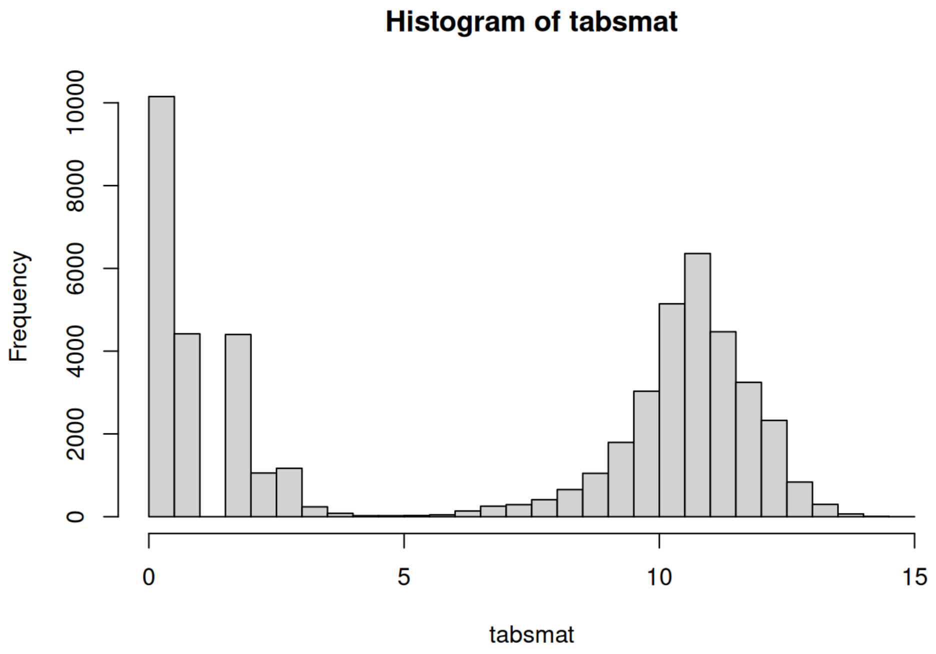
Normalized read count distribution of all samples

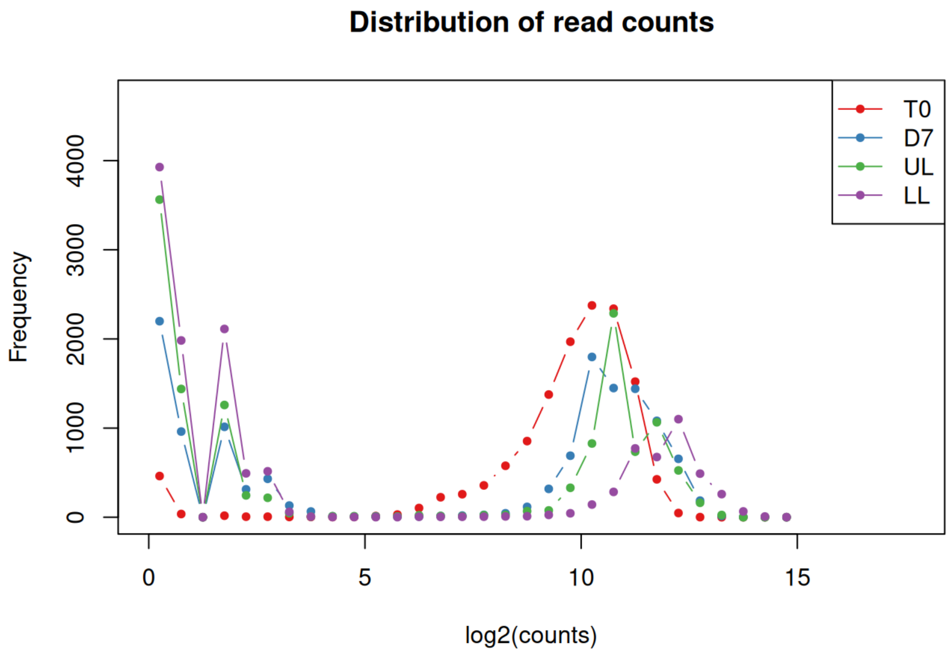
The following figure shows the distribution of median-normalized read counts in all samples.





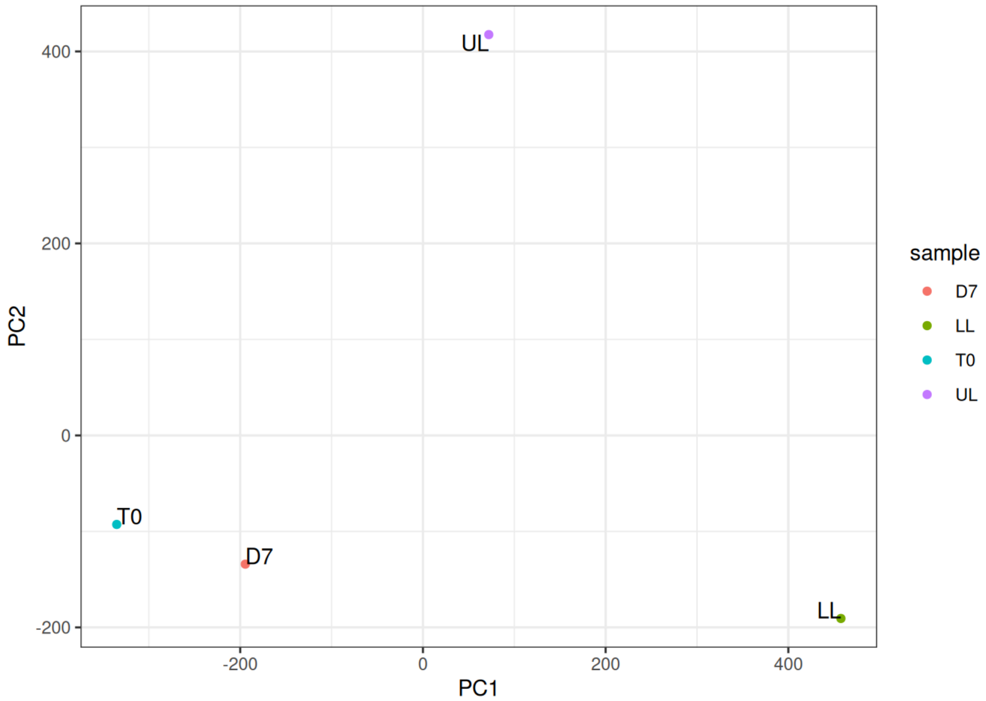
The following figure shows the histogram of median-normalized read counts in all samples.

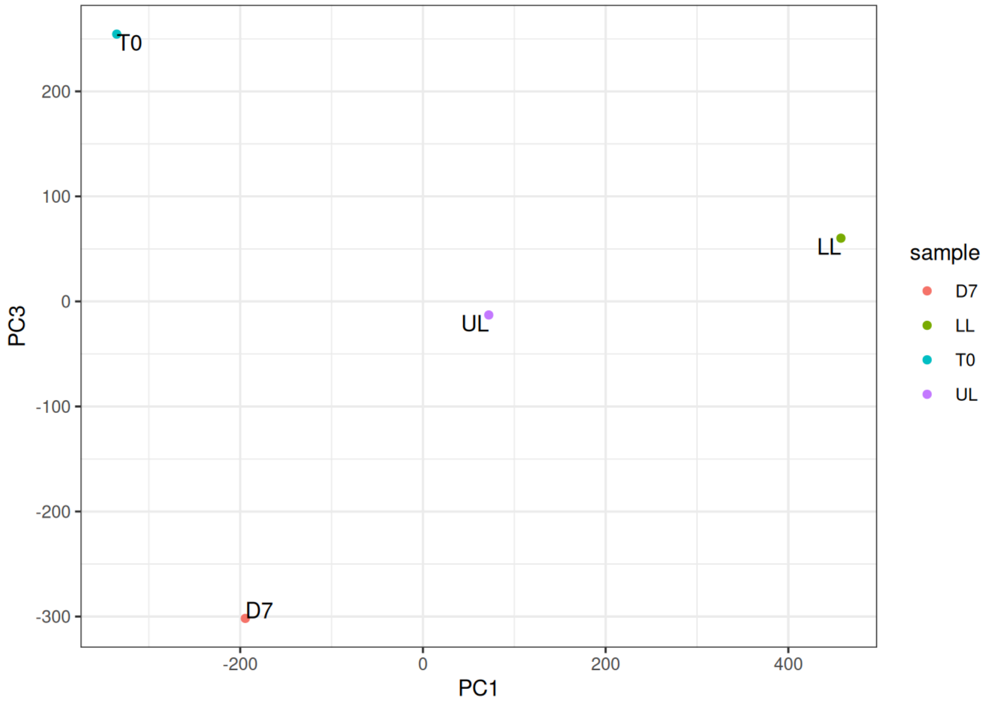


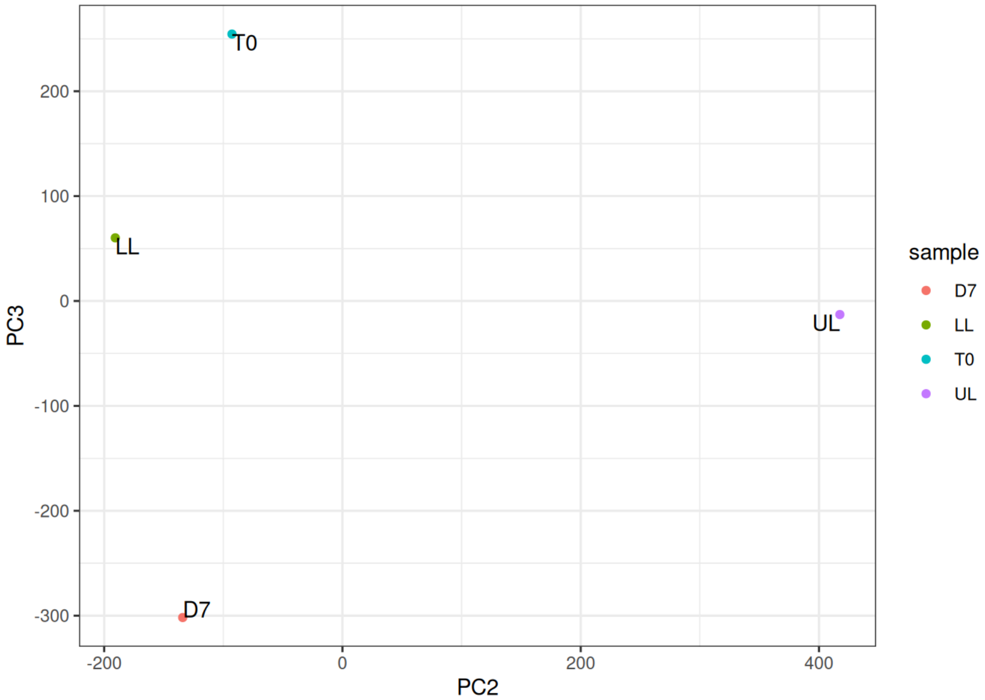


Principle Component Analysis

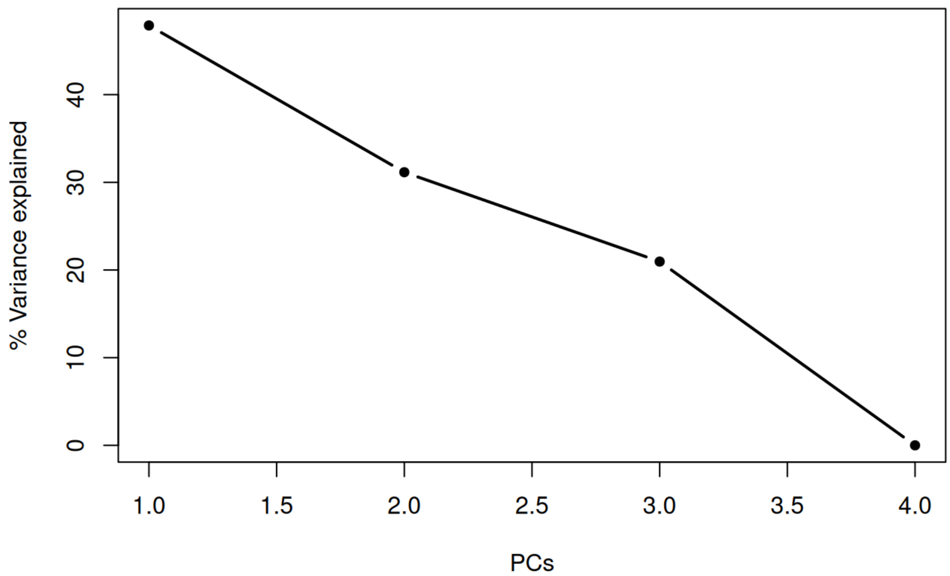
The following figure shows the first 2 principle components (PCs) from the Principle Component Analysis (PCA), and the percentage of variances explained by the top PCs.


PC2
400
200
0
-200
UL
T0
D7
LL
-200
0
200
400
PC1
sample
D7
LL
T0
UL


PC3
200
100
0
-100
-200
-300
T0
LL
UL
D7
-200
0
200
400
PC1
sample
D7
LL
T0
UL

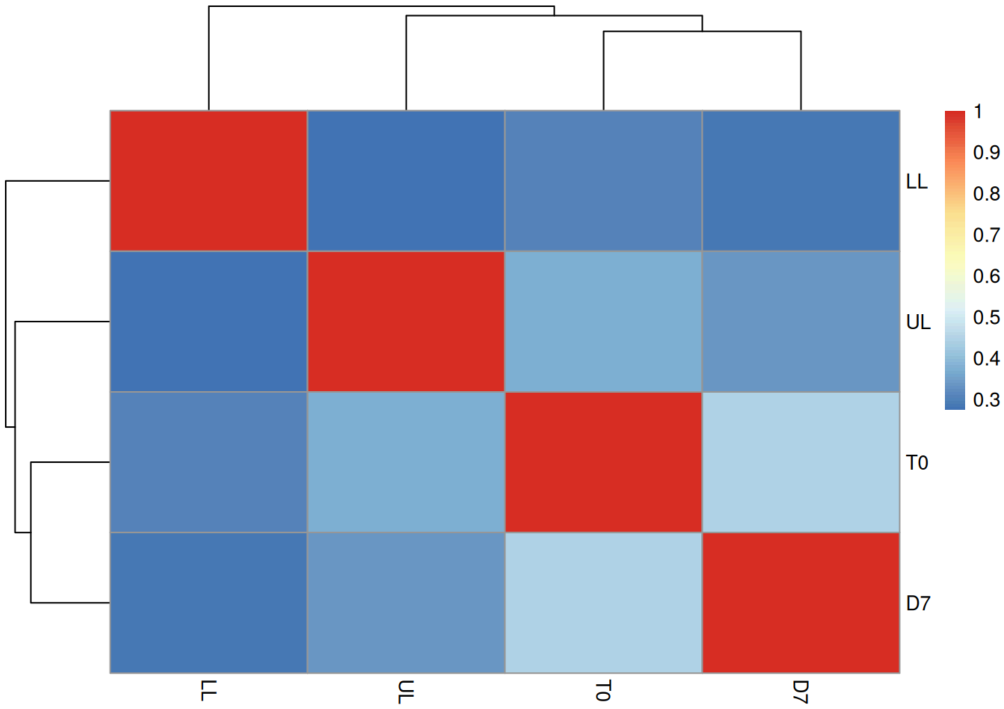


The variance of the PCs



Sample clustering

The following figure shows the sample clustering result.



LL
UL
T0
D7
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
LL
UL
T0
D7